Final Project Report

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PW SKILLS

FINAL PROJECT REPORT

Project Title

Cryptocurrency Historical Prices Prediction using Machine Learning

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Course:
Data Analytics with Machine Learning

Organization:
PW Skills

Abstract

This project focuses on predicting cryptocurrency market volatility using Machine Learning techniques. Historical cryptocurrency price data was collected, preprocessed, analyzed, and used to train a Random Forest Regression model. The project includes data preprocessing, exploratory data analysis, feature engineering, model training, evaluation, and prediction generation. The developed model provides useful insights into cryptocurrency price movements and can assist traders and investors in understanding market volatility.

Project Objective

The objective of this project is to build a machine learning model capable of predicting cryptocurrency volatility using historical market data. The project aims to improve decision-making by analyzing price trends and generating reliable predictions.

Dataset Description

The project uses the Cryptocurrency Historical Prices Dataset containing historical market information including Open, High, Low, Close prices, Volume, Market Capitalization, Date, Timestamp, and Cryptocurrency Name.

Dataset Size:
72,946 records

Total Features:
10 columns

Methodology

The project was completed using the following methodology:

• Data Collection
• Data Cleaning
• Exploratory Data Analysis (EDA)
• Feature Engineering
• Feature Scaling
• Train-Test Split
• Model Training using Random Forest Regressor
• Model Evaluation
• Prediction Generation
• Model Saving

Tools & Technologies

Programming Language:
Python

Development Environment:
Google Colab

Libraries:
Pandas
NumPy
Matplotlib

Seaborn
Scikit-learn
Joblib

Results

The machine learning model successfully learned patterns from historical cryptocurrency data and generated volatility predictions. Feature engineering improved prediction quality, while preprocessing ensured clean and consistent input data. The Random Forest Regression model demonstrated effective performance for the given dataset.

Challenges

During the project, challenges such as dataset preprocessing, handling infinite values, feature engineering, and model training were encountered. These issues were resolved through appropriate preprocessing techniques and data validation.

Future Scope

Future improvements may include:

• Deep Learning models
• LSTM-based time series forecasting
• Real-time cryptocurrency prediction
• Streamlit web application deployment
• Live market data integration

Conclusion

The Cryptocurrency Historical Prices Prediction project demonstrates how machine learning can be applied to financial market analysis. Through data preprocessing, exploratory analysis, feature engineering, and model development, an effective prediction system was created. This project provides a strong foundation for future enhancements involving real-time prediction and advanced machine learning techniques.

References

- PW Skills Course Material
- Scikit-learn Documentation
- Pandas Documentation
- NumPy Documentation
- Matplotlib Documentation